Water quality analyses for selected local authority areas (2020-2025)

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Table of contents


```r
options(repos = c(CRAN = "https://cloud.r-project.org/"))
library(tidyverse)
library(rio)
library(readr)
library(patchwork)
```

Data source: https://environment.data.gov.uk/water-quality

Two pairs of geographically proximate sampling sites in the south of England were selected for analysis:

```
-   E06000035 Medway
-   E07000064 Rother
-   E06000046 Isle of Wight
-   E07000091 New Forest
```

Datasets

```r
# Import from csv
med <- rio::import("data/E06000035_Medway.csv")
rot <- rio::import("data/E07000064_Rother.csv")
iow <- rio::import("data/E06000046_Isle_of_Wight.csv")
nfo <- rio::import("data/E07000091_New_Forest.csv")

# Add location column
med$location <- "Medway"
rot$location <- "Rother"
iow$location <- "Isle_of_Wight"
nfo$location <- "New_Forest"
```

Reusable function

```r
plot_fun <- function(df, var) {

  # correct n (excl. missing values).
  n <- sum(!is.na(df[[var]]))

  # 1. Histogram (distribution)
  p1 <- ggplot(df, aes(x = .data[[var]])) +
    geom_histogram(bins = 30) +
    theme_bw() +
    ggtitle(paste0("Hist. of ", var))

  # 2. Overall boxplot (outliers)
  p2 <- ggplot(df, aes(y = .data[[var]])) +
    geom_boxplot() +
    theme_bw() +
    ggtitle(paste0("Boxplot of ", var))
```

```
  # monthly trend (sesonality)
  p3 <- ggplot(df, aes(x = phenomenonTime, y = .data[[var]])) +
    geom_line(alpha = 0.5) +
    geom_smooth() +
    theme_bw() +
    ggtitle(paste0("Time Series of ", var))

  # yearly trend
  p4 <- ggplot(df, aes(x = YEAR, y = .data[[var]])) +
    stat_summary(fun = mean, geom = "line") +
    stat_summary(fun = mean, geom = "point") +
    theme_bw() +
    ggtitle(paste0("Yearly Mean of ", var))

  # monthly distribution
  p5 <- ggplot(df, aes(x = factor(MONTH), y = .data[[var]])) +
    geom_boxplot() +
    theme_bw() +
    ggtitle(paste0("Monthly Dist. of ", var))

  # yearly distribution
  p6 <- ggplot(df, aes(x = factor(YEAR), y = .data[[var]])) +
    geom_boxplot() +
    ggtitle(paste0("Yearly Dist of ", var))

  # layout
  (p1 | p2) /
  (p3 | p4) /
  (p5 | p6) +
    plot_annotation(
      title = paste0("Exploratory Analysis of ", var, " (n = ", n, ")")
    )
}
```

Inspect datasets

E06000035_Medway

```
# ------ 1. Data cleaning and quality check ------
```

```r
# Specify the dataset
df <- med # <- only change this

# categorical nominal variables as factors
df$location <- as.factor(df$location)
df$samplingPoint.notation <- as.factor(df$samplingPoint.notation)
df$samplingPoint.prefLabel <- as.factor(df$samplingPoint.prefLabel)
df$samplingPoint.region <- as.factor(df$samplingPoint.region)
df$samplingPoint.area <- as.factor(df$samplingPoint.area)
df$samplingPoint.subArea <- as.factor(df$samplingPoint.subArea)
df$samplingPoint.samplingPointStatus <- as.factor(df$samplingPoint.samplingPointStatus)
df$samplingPoint.samplingPointType <- as.factor(df$samplingPoint.samplingPointType)
df$samplingPurpose <- as.factor(df$samplingPurpose)
df$sampleMaterialType <- as.factor(df$sampleMaterialType)

# create month end year
df <- df %>%
  mutate(
    YEAR = year(phenomenonTime),
    MONTH = month(phenomenonTime)
  )

# check for missing data
colSums(is.na(df))
```

```
                            Date           samplingPoint.notation
                               0                                0
         samplingPoint.prefLabel           samplingPoint.easting
                               0                                0
          samplingPoint.northing            samplingPoint.region
                               0                                0
              samplingPoint.area           samplingPoint.subArea
                               0                                0
samplingPoint.samplingPointStatus   samplingPoint.samplingPointType
                               0                                0
                   phenomenonTime                  samplingPurpose
                               0                                0
              sampleMaterialType                         pH Water
                               0                               22
                       Temp Water                       Cond @ 25C
                             147                               19
                         location                             YEAR
```

```
                                   0                                   0
                               MONTH
                                   0
```

```r
# remove duplicates, if any
df <- df %>%
  distinct()

# check structure
str(df)
```

```
'data.frame':   215 obs. of  19 variables:
 $ Date                              : IDate, format: "2020-01-10" "2020-02-
25" ...
 $ samplingPoint.notation            : Factor w/ 8 levels "SO-5GWQ0462",..: 4 4 4 4 4 4 4 4 4
 $ samplingPoint.prefLabel           : Factor w/ 8 levels "ALLENS HILL INVESTIGATION AT CLIFF
 $ samplingPoint.easting             : int  572949 572949 572949 572949 572949 572949 572949 5
 $ samplingPoint.northing            : int  176722 176722 176722 176722 176722 176722 176722 1
 $ samplingPoint.region              : Factor w/ 1 level "Southern": 1 1 1 1 1 1 1 1 1 1 ...
 $ samplingPoint.area                : Factor w/ 1 level "SOUTHERN - KENT AND SOUTH LONDON": 1
 $ samplingPoint.subArea             : Factor w/ 1 level "L&W KENT AND SOUTH LONDON (SO)": 1 1
 $ samplingPoint.samplingPointStatus: Factor w/ 1 level "OPEN": 1 1 1 1 1 1 1 1 1 1 ...
 $ samplingPoint.samplingPointType  : Factor w/ 5 levels "FRESHWATER - LAKES/PONDS/RESERVOIRS
 $ phenomenonTime                    : POSIXct, format: "2020-01-10 09:38:00" "2020-
02-25 12:01:00" ...
 $ samplingPurpose                   : Factor w/ 4 levels "COMPLIANCE AUDIT (PERMIT)",..: 1 1
 $ sampleMaterialType                : Factor w/ 4 levels "ESTUARINE WATER",..: 1 1 1 1 1 1 1
 $ pH Water                          : num  8.45 8.6 8.63 8.43 8.74 8.67 8.61 8.69 8.83 8.66
 $ Temp Water                        : num  NA NA NA NA NA NA NA NA NA NA ...
 $ Cond @ 25C                        : int  44870 40656 38052 41183 40625 41310 39550 38616 37
 $ location                          : Factor w/ 1 level "Medway": 1 1 1 1 1 1 1 1 1 1 ...
 $ YEAR                              : num  2020 2020 2020 2021 2021 ...
 $ MONTH                             : num  1 2 3 10 11 12 1 2 3 5 ...
```

```r
# check ranges
summary(df)
```

```
      Date            samplingPoint.notation
 Min.   :2020-01-10   SO-E0007222:47
 1st Qu.:2022-03-17   SO-E0007223:46
 Median :2023-11-21   SO-E0007220:44
```

```
 Mean   :2023-08-01   SO-E0004880:36
 3rd Qu.:2025-01-27   SO-E0007235:21
 Max.   :2025-12-16   SO-E0007331:16
                      (Other)    : 5
                             samplingPoint.prefLabel samplingPoint.easting
 ALLENS HILL INVESTIGATION AT CLIFFE POOLS:47         Min.   :570951
 CLIFFE POOL CAR PARK                     :46         1st Qu.:572261
 ALPHA LAKE OUTLET                        :44         Median :572307
 CLIFFE QUARRY, CLIFFE, KENT              :36         Mean   :572918
 BUCKLAND LAKE SOUTH                      :21         3rd Qu.:572949
 DOWNSTREAM KINGSNORTH INDUSTRIAL         :16         Max.   :581238
 (Other)                                  : 5
 samplingPoint.northing samplingPoint.region
 Min.   :173438         Southern:215
 1st Qu.:175885
 Median :175988
 Mean   :175912
 3rd Qu.:176041
 Max.   :176782

                       samplingPoint.area
 SOUTHERN - KENT AND SOUTH LONDON:215




                   samplingPoint.subArea samplingPoint.samplingPointStatus
 L&W KENT AND SOUTH LONDON (SO):215       OPEN:215




                                                       samplingPoint.samplingPointType
 FRESHWATER - LAKES/PONDS/RESERVOIRS                                          : 36
 GROUNDWATER - BOREHOLE                                                       :  2
 GROUNDWATER - WELLS & ADITS                                                  :  3
 TRADE DISCHARGES - SITE DRAINAGE (CONTAM SURFACE WATER, NOT WASTE SIT: 16
 WASTE SITE - INDUSTRIAL LANDFILL                                             :158
```

```
 phenomenonTime
 Min.   :2020-01-10 09:38:00
 1st Qu.:2022-03-17 10:48:00
 Median :2023-11-21 10:30:00
 Mean   :2023-08-01 18:40:22
 3rd Qu.:2025-01-27 10:43:30
 Max.   :2025-12-16 12:31:00

                                          samplingPurpose
 COMPLIANCE AUDIT (PERMIT)                        :137
 ENVIRONMENTAL MONITORING STATUTORY (EU DIRECTIVES): 22
 PLANNED INVESTIGATION (LOCAL MONITORING)          : 21
 PLANNED INVESTIGATION (NATIONAL AGENCY POLICY)    : 35



                    sampleMaterialType    pH Water       Temp Water
 ESTUARINE WATER                 :137    Min.   :6.950   Min.   : 3.20
 GROUNDWATER                     :  5    1st Qu.:8.300   1st Qu.: 7.70
 POND / LAKE / RESERVOIR WATER: 57       Median :8.520   Median :11.15
 RIVER / RUNNING SURFACE WATER: 16       Mean   :8.418   Mean   :12.19
                                         3rd Qu.:8.660   3rd Qu.:16.62
                                         Max.   :9.480   Max.   :23.70
                                         NA's   :22      NA's   :147
   Cond @ 25C       location        YEAR          MONTH
 Min.   :  1168   Medway:215   Min.   :2020   Min.   : 1.000
 1st Qu.:  2100                1st Qu.:2022   1st Qu.: 3.000
 Median : 32000                Median :2023   Median : 6.000
 Mean   : 28192                Mean   :2023   Mean   : 6.386
 3rd Qu.: 40602                3rd Qu.:2025   3rd Qu.:10.000
 Max.   :377705                Max.   :2025   Max.   :12.000
 NA's   :19
```

```r
# pH of Water
plot_fun(df, "pH Water")
```

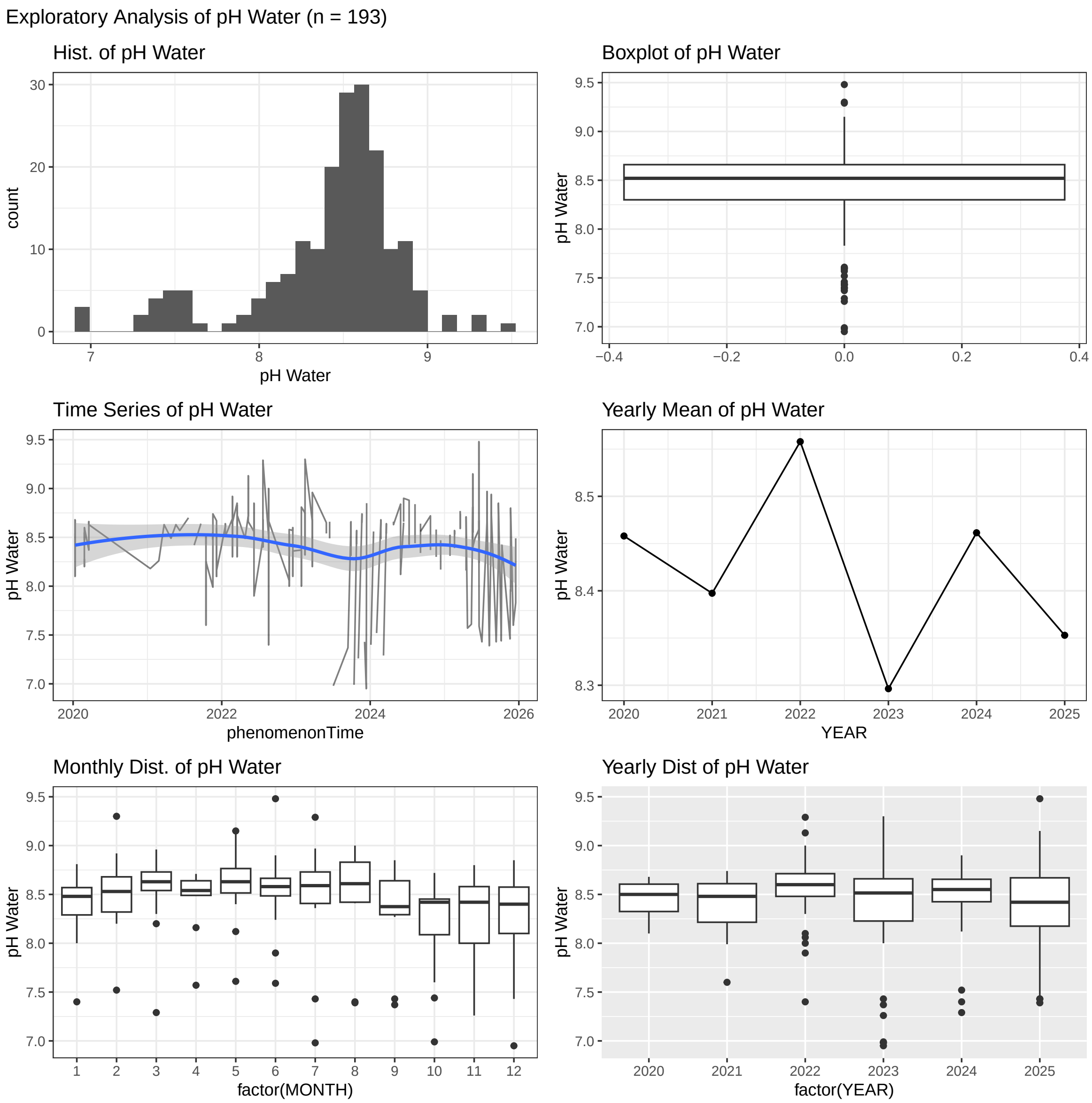

Figure 1: E06000035 Medway. pH (2020-2025)

```
# Temperature of Water
plot_fun(df, "Temp Water")
```

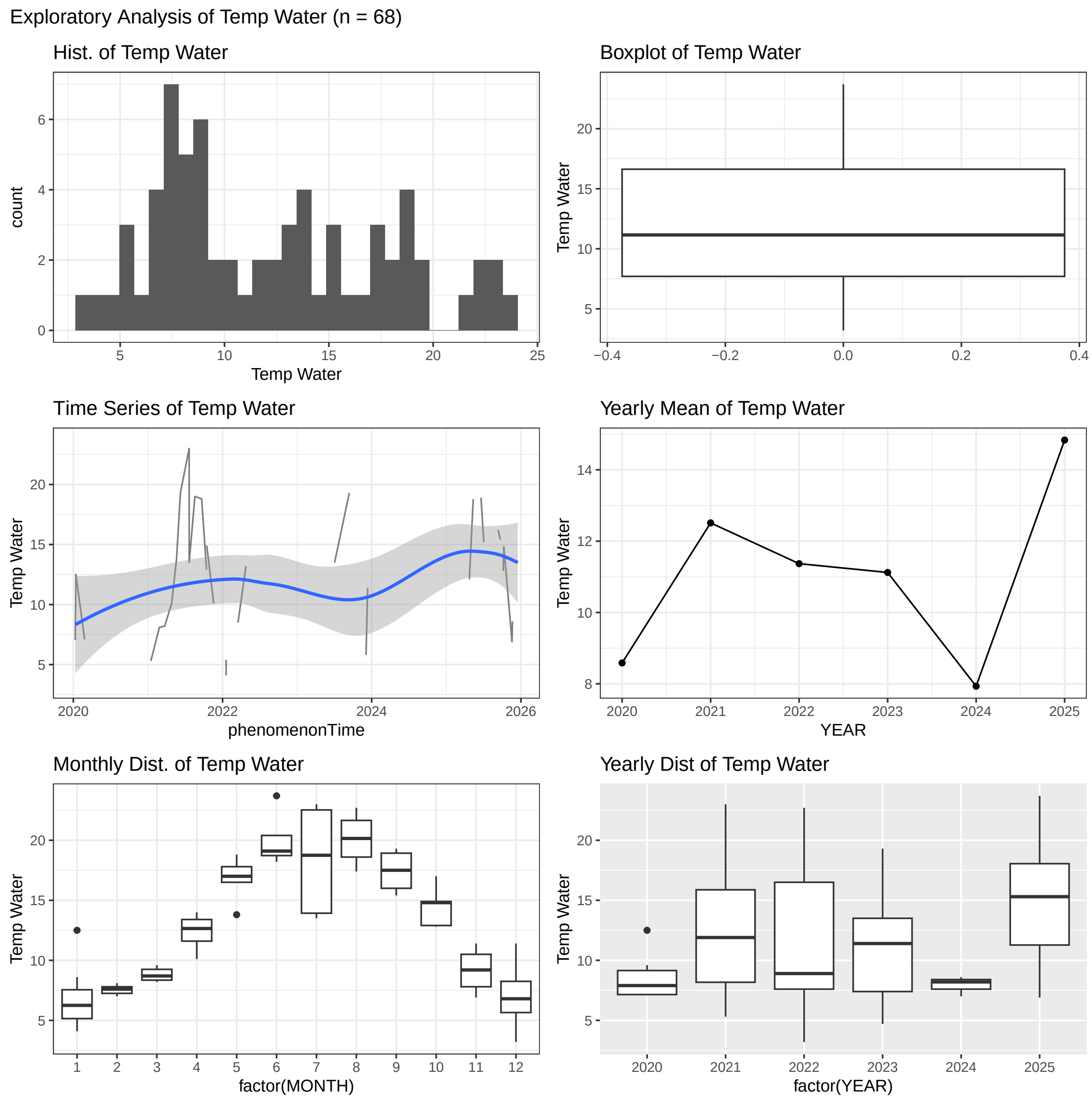

Figure 2: E06000035 Medway. Temperature (2020-2025)

```
# Conductivity @ 25 degrees
plot_fun(df, "Cond @ 25C")
```

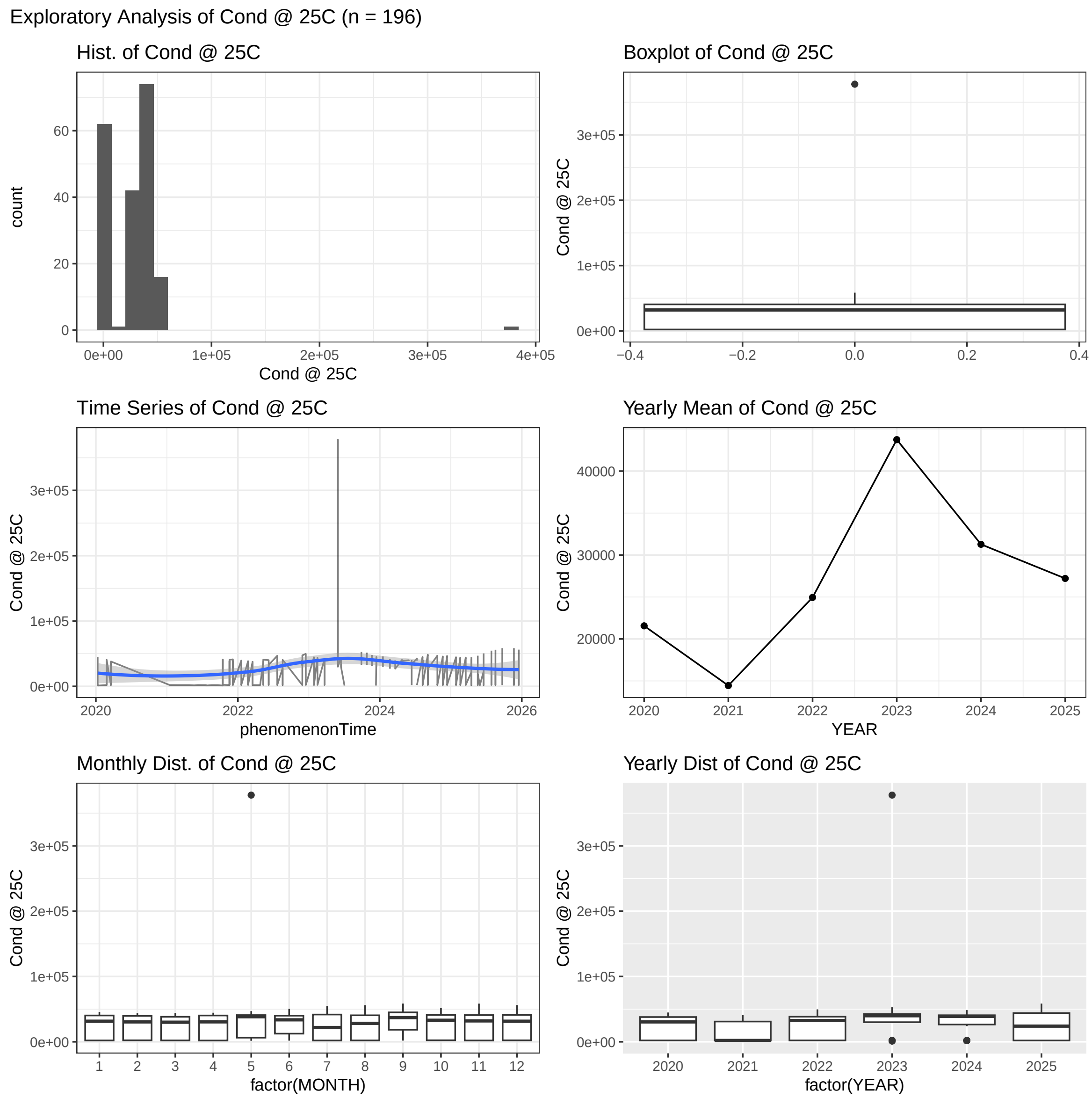

Figure 3: E06000035 Medway. Conductivity @ 25C (2020-2025)

E07000064_Rother

```
# ------ 1. Data cleaning and quality check ------

# Specify the dataset
```

```r
df <- rot # <- only change this

# categorical nominal variables as factors
df$location <- as.factor(df$location)
df$samplingPoint.notation <- as.factor(df$samplingPoint.notation)
df$samplingPoint.prefLabel <- as.factor(df$samplingPoint.prefLabel)
df$samplingPoint.region <- as.factor(df$samplingPoint.region)
df$samplingPoint.area <- as.factor(df$samplingPoint.area)
df$samplingPoint.subArea <- as.factor(df$samplingPoint.subArea)
df$samplingPoint.samplingPointStatus <- as.factor(df$samplingPoint.samplingPointStatus)
df$samplingPoint.samplingPointType <- as.factor(df$samplingPoint.samplingPointType)
df$samplingPurpose <- as.factor(df$samplingPurpose)
df$sampleMaterialType <- as.factor(df$sampleMaterialType)

# create month end year
df <- df %>%
  mutate(
    YEAR = year(phenomenonTime),
    MONTH = month(phenomenonTime)
  )

# check for missing data
colSums(is.na(df))
```

```
                             Date           samplingPoint.notation
                                0                                0
          samplingPoint.prefLabel            samplingPoint.easting
                                0                                0
           samplingPoint.northing             samplingPoint.region
                                0                                0
               samplingPoint.area            samplingPoint.subArea
                                0                                0
samplingPoint.samplingPointStatus   samplingPoint.samplingPointType
                                0                                0
                   phenomenonTime                  samplingPurpose
                                0                                0
               sampleMaterialType                         pH Water
                                0                              107
                       Temp Water                      Cond @ 25C
                              527                              558
                         location                             YEAR
                                0                                0
```

```
                  MONTH
                      0
```

```r
# remove duplicates, if any
df <- df %>%
  distinct()

# check structure
str(df)
```

```
'data.frame':   1934 obs. of  19 variables:
 $ Date                             : IDate, format: "2020-01-09" "2020-02-
09" ...
 $ samplingPoint.notation           : Factor w/ 82 levels "SO-5GWQ0488",..: 17 17 17 17 17 17
 $ samplingPoint.prefLabel          : Factor w/ 82 levels "BATTLE SEWAGE TREATMENT WORKS FINA
 $ samplingPoint.easting            : int  575970 575970 575970 575970 575970 575970 575970 5
 $ samplingPoint.northing           : int  116160 116160 116160 116160 116160 116160 116160 1
 $ samplingPoint.region             : Factor w/ 2 levels "Southern","Thames": 1 1 1 1 1 1 1 1
 $ samplingPoint.area               : Factor w/ 3 levels "SOUTHERN - KENT AND SOUTH LONDON",
 $ samplingPoint.subArea            : Factor w/ 4 levels "L&W EAST - SSD",..: 2 2 2 2 2 2 2 2
 $ samplingPoint.samplingPointStatus: Factor w/ 1 level "OPEN": 1 1 1 1 1 1 1 1 1 1 ...
 $ samplingPoint.samplingPointType  : Factor w/ 10 levels "FRESHWATER - LAKES/PONDS/RESERVOI
 $ phenomenonTime                   : POSIXct, format: "2020-01-09 10:50:00" "2020-
02-09 11:25:00" ...
 $ samplingPurpose                  : Factor w/ 10 levels "COMPLIANCE AUDIT (PERMIT)",..: 10
 $ sampleMaterialType               : Factor w/ 6 levels "ANY TRADE EFFLUENT",..: 2 2 2 2 2 2
 $ pH Water                         : num  7.32 7.5 7.84 7.33 7.49 7.64 7.5 7.81 7.37 7.43 .
 $ Temp Water                       : num  NA NA NA NA NA NA NA NA NA NA ...
 $ Cond @ 25C                       : int  NA NA NA NA NA NA NA NA NA NA ...
 $ location                         : Factor w/ 1 level "Rother": 1 1 1 1 1 1 1 1 1 1 ...
 $ YEAR                             : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020
 $ MONTH                            : num  1 2 3 4 5 6 8 9 10 11 ...
```

```r
# check ranges
summary(df)
```

```
      Date              samplingPoint.notation
 Min.   :2020-01-03   SO-E0001621:  81
 1st Qu.:2022-05-12   SO-F0001970:  79
 Median :2023-10-21   SO-E0001608:  70
 Mean   :2023-07-30   SO-E0000109:  65
```

```
3rd Qu.:2024-12-16   SO-E0001735:  60
Max.   :2025-12-18   SO-E0001604:  59
                     (Other)    :1520

WESTFIELD SEWAGE TREATMENT WORKS FINAL EFFLUENT SAMPLED AT OUTFALL
CATSFIELD SKINNERS LANE SEWAGE TREATMENT WORKS FINAL EFFLUENT SAMPLED AT OUTFALL (STORM CON
GUESTLING SEWAGE TREATMENT WORKS FINAL EFFLUENT SAMPLED AT OUTFALL
CAMBER SANDS EC BATHING WATER (13900)
KENT DITCH AT A229 DOWNSTREAM CONFLUENCE HAWKHURST SOUTH STREAM
MARSHAM SEWER DOWNSTREAM CONFLUENCE WITH TRIBUTARIES AT CLIFFEND
(Other)
samplingPoint.easting samplingPoint.northing samplingPoint.region
Min.   :565069        Min.   :106232         Southern:1903
1st Qu.:571698        1st Qu.:115385         Thames  :  31
Median :579400        Median :118473
Mean   :579733        Mean   :119050
3rd Qu.:585890        3rd Qu.:122606
Max.   :600387        Max.   :133731

                        samplingPoint.area
SOUTHERN - KENT AND SOUTH LONDON :1658
SOUTHERN - SOLENT AND SOUTH DOWNS: 245
THAMES - KENT AND SOUTH LONDON   :  31




                   samplingPoint.subArea samplingPoint.samplingPointStatus
L&W EAST - SSD                : 20      OPEN:1934
L&W KENT AND SOUTH LONDON (SO):1658
L&W KENT AND SOUTH LONDON (TH):  31
L&W SOUTH - SSD               : 225




                                          samplingPoint.samplingPointType
FRESHWATER - RIVERS                                     :1191
SEWAGE DISCHARGES - FINAL/TREATED EFFLUENT - WATER COMPANY: 333
FRESHWATER - UNSPECIFIED                                : 151
FRESHWATER - LAKES/PONDS/RESERVOIRS                     :  73
SALINE WATER - DESIGNATED BATHING BEACHES               :  65
MISCELLANEOUS DISCHARGES - MINE/GROUNDWATER AS RAISED   :  50
(Other)                                                 :  71
```

```
 phenomenonTime
 Min.   :2020-01-03 10:00:00
 1st Qu.:2022-05-12 11:27:45
 Median :2023-10-21 09:49:30
 Mean   :2023-07-30 18:38:14
 3rd Qu.:2024-12-16 10:17:00
 Max.   :2025-12-18 15:55:00

                                              samplingPurpose
 MONITORING  (NATIONAL AGENCY POLICY)                 :802
 ENVIRONMENTAL MONITORING STATUTORY (EU DIRECTIVES)    :680
 WATER QUALITY OPERATOR SELF MONITORING COMPLIANCE DATA:382
 COMPLIANCE AUDIT (PERMIT)                             : 51
 PLANNED INVESTIGATION (NATIONAL AGENCY POLICY)        : 13
 PLANNED INVESTIGATION (LOCAL MONITORING)              :  2
 (Other)                                               :  4
                     sampleMaterialType    pH Water       Temp Water
 ANY TRADE EFFLUENT            :  50     Min.   :5.200   Min.   : 2.40
 FINAL SEWAGE EFFLUENT         : 383     1st Qu.:7.290   1st Qu.: 8.50
 GROUNDWATER                   :  20     Median :7.500   Median :11.70
 POND / LAKE / RESERVOIR WATER:  64     Mean   :7.513   Mean   :11.94
 RIVER / RUNNING SURFACE WATER:1352     3rd Qu.:7.700   3rd Qu.:15.30
 SEA WATER                     :  65     Max.   :9.400   Max.   :24.40
                                         NA's   :107     NA's   :527
   Cond @ 25C        location         YEAR          MONTH
 Min.   :  118.0   Rother:1934   Min.   :2020   Min.   : 1.000
 1st Qu.:  335.0                 1st Qu.:2022   1st Qu.: 4.000
 Median :  426.0                 Median :2023   Median : 7.000
 Mean   : 1016.3                 Mean   :2023   Mean   : 6.643
 3rd Qu.:  593.2                 3rd Qu.:2024   3rd Qu.:10.000
 Max.   :45875.0                 Max.   :2025   Max.   :12.000
 NA's   :558
```

```r
# pH of Water
plot_fun(df, "pH Water")
```

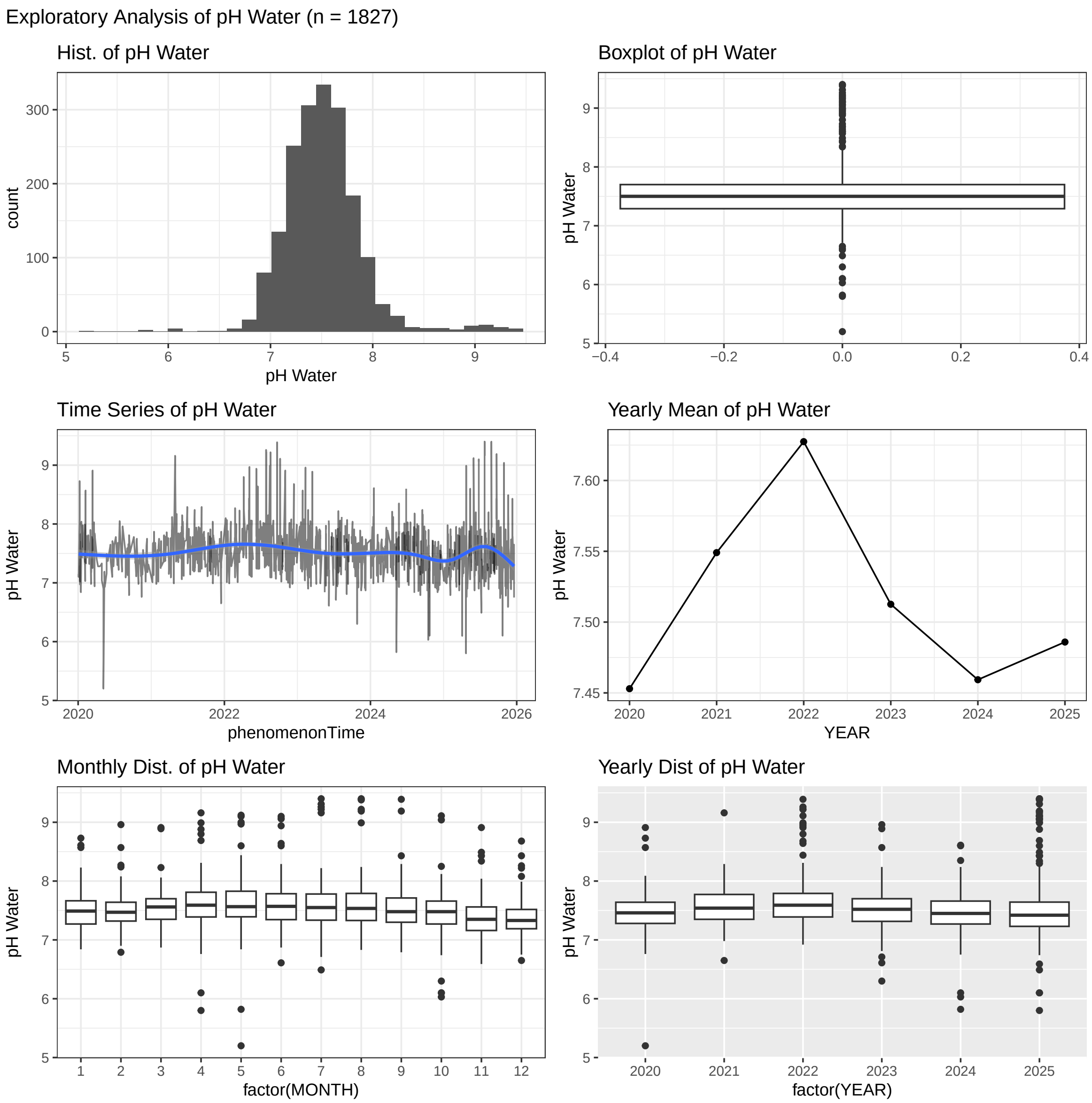

Figure 4: E07000064 Rother. pH (2020-2025)

```
# Temperature of Water
plot_fun(df, "Temp Water")
```

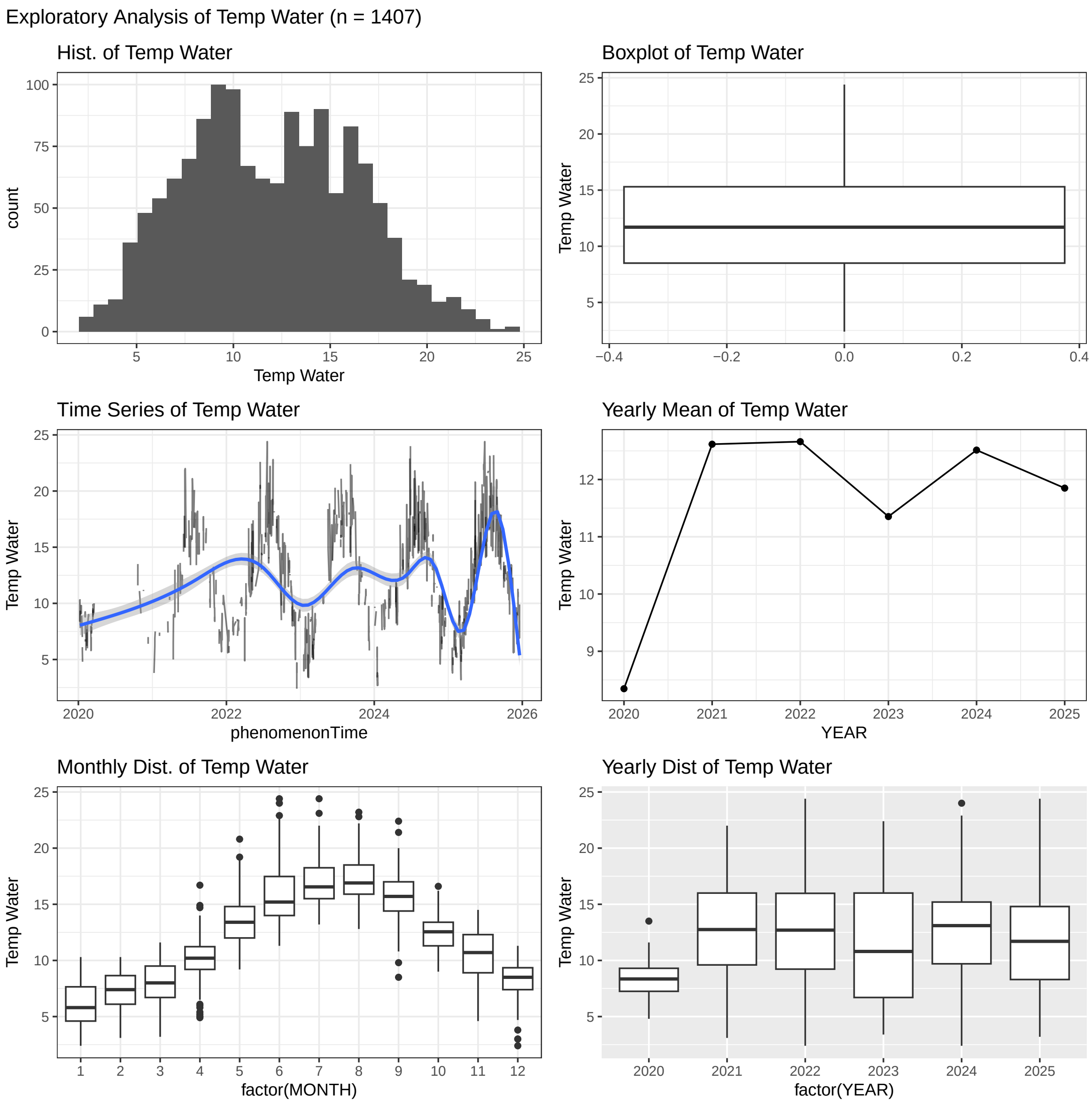

Figure 5: E07000064 Rother. Temperature (2020-2025)

```
# Conductivity @ 25 degrees
plot_fun(df, "Cond @ 25C")
```

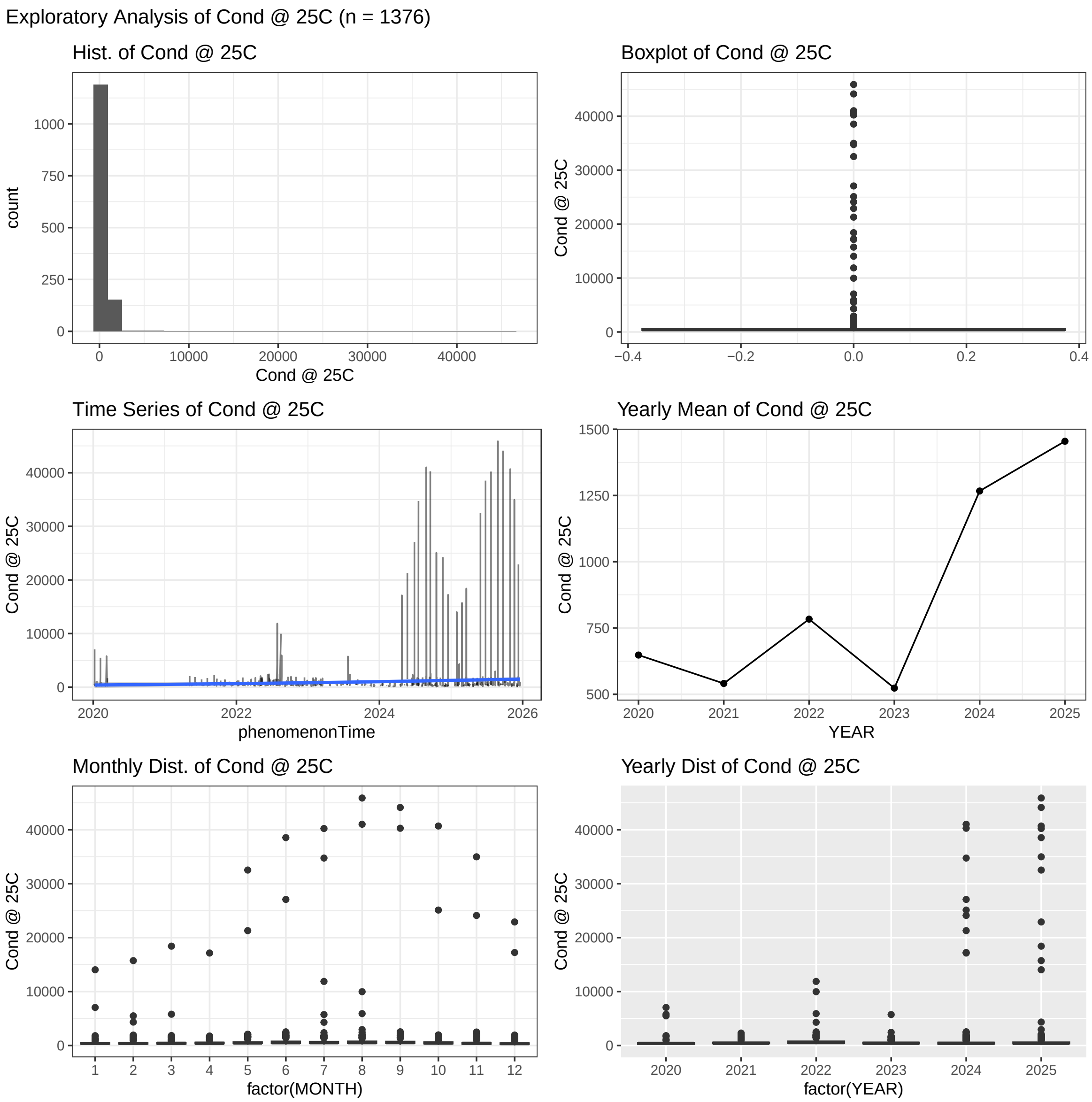

Figure 6: E07000064 Rother. Conductivity @ 25C (2020-2025)

E06000046_Isle_of_Wight

```
# ------ 1. Data cleaning and quality check ------

# Specify the dataset
```

```r
df <- iow # <- only change this

# categorical nominal variables as factors
df$location <- as.factor(df$location)
df$samplingPoint.notation <- as.factor(df$samplingPoint.notation)
df$samplingPoint.prefLabel <- as.factor(df$samplingPoint.prefLabel)
df$samplingPoint.region <- as.factor(df$samplingPoint.region)
df$samplingPoint.area <- as.factor(df$samplingPoint.area)
df$samplingPoint.subArea <- as.factor(df$samplingPoint.subArea)
df$samplingPoint.samplingPointStatus <- as.factor(df$samplingPoint.samplingPointStatus)
df$samplingPoint.samplingPointType <- as.factor(df$samplingPoint.samplingPointType)
df$samplingPurpose <- as.factor(df$samplingPurpose)
df$sampleMaterialType <- as.factor(df$sampleMaterialType)

# create month end year
df <- df %>%
  mutate(
    YEAR = year(phenomenonTime),
    MONTH = month(phenomenonTime)
  )

# check for missing data
colSums(is.na(df))
```

```
                             Date            samplingPoint.notation
                                0                                 0
          samplingPoint.prefLabel             samplingPoint.easting
                                0                                 0
           samplingPoint.northing              samplingPoint.region
                                0                                 0
               samplingPoint.area             samplingPoint.subArea
                                0                                 0
samplingPoint.samplingPointStatus   samplingPoint.samplingPointType
                                0                                 0
                   phenomenonTime                   samplingPurpose
                                0                                 0
               sampleMaterialType                          pH Water
                                0                               436
                       Temp Water                       Cond @ 25C
                               25                               447
                         location                              YEAR
                                0                                 0
```

```
                                           MONTH
                                               0
```

```r
# remove duplicates, if any
df <- df %>%
  distinct()

# check structure
str(df)
```

```
'data.frame':   1968 obs. of  19 variables:
 $ Date                               : IDate, format: "2020-01-23" "2020-02-
11" ...
 $ samplingPoint.notation             : Factor w/ 64 levels "SO-5GWQ0480",..: 29 29 29 29 29 29
 $ samplingPoint.prefLabel            : Factor w/ 64 levels "APSE NEW BARN FARM, NEWCHURCH, IOW
 $ samplingPoint.easting              : int  450200 450200 450200 450200 450200 450200 450200 4
 $ samplingPoint.northing             : int  89500 89500 89500 89500 89500 89500 89500 89500 89
 $ samplingPoint.region               : Factor w/ 1 level "Southern": 1 1 1 1 1 1 1 1 1 1 ...
 $ samplingPoint.area                 : Factor w/ 1 level "SOUTHERN - SOLENT AND SOUTH DOWNS":
 $ samplingPoint.subArea              : Factor w/ 2 levels "L&W EAST - SSD",..: 1 1 1 1 1 1 1 1
 $ samplingPoint.samplingPointStatus: Factor w/ 2 levels "CLOSED","OPEN": 2 2 2 2 2 2 2 2 2 2
 $ samplingPoint.samplingPointType  : Factor w/ 7 levels "FRESHWATER - RIVERS",..: 7 7 7 7 7
 $ phenomenonTime                     : POSIXct, format: "2020-01-23 10:35:00" "2020-
02-11 10:01:00" ...
 $ samplingPurpose                    : Factor w/ 7 levels "ENVIRONMENTAL MONITORING STATUTORY
 $ sampleMaterialType                 : Factor w/ 4 levels "ESTUARINE WATER",..: 1 1 1 1 1 1 1
 $ pH Water                           : num  NA NA NA NA NA NA NA NA NA NA ...
 $ Temp Water                         : num  8.3 7 7.8 6.4 8.9 13.1 16.1 17.3 7.4 7.9 ...
 $ Cond @ 25C                         : num  NA NA NA NA NA NA NA NA NA NA ...
 $ location                           : Factor w/ 1 level "Isle_of_Wight": 1 1 1 1 1 1 1 1 1 1
 $ YEAR                               : num  2020 2020 2020 2021 2021 ...
 $ MONTH                              : num  1 2 12 1 2 5 7 9 12 1 ...
```

```r
# check ranges
summary(df)
```

```
      Date               samplingPoint.notation
 Min.   :2020-01-07   SO-Y0004344:  66
 1st Qu.:2022-09-14   SO-Y0004300:  63
 Median :2023-12-13   SO-Y0004283:  61
 Mean   :2023-09-24   SO-Y0004347:  61
```

```
3rd Qu.:2024-12-06   SO-Y0004399:  59
Max.   :2025-12-19   SO-Y0004402:  59
                     (Other)    :1599
                              samplingPoint.prefLabel samplingPoint.easting
RODGE BROOK NORTH WEST OF PORCHFIELD      :  66       Min.   :432195
SHANKLIN EC BATHING WATER (18500)         :  63       1st Qu.:442065
CAULE BOURNE AT SHALFLEET                 :  61       Median :450355
RYDE EAST SANDS (BEACH MONITORING) (17900):  61       Mean   :449727
R EASTERN YAR AT WEED GRID ST HELENS      :  59       3rd Qu.:456859
RIVER EASTERN YAR AT BURNT HOUSE          :  59       Max.   :463755
(Other)                                   :1599
samplingPoint.northing samplingPoint.region
Min.   :77341          Southern:1968
1st Qu.:83365
Median :86900
Mean   :86385
3rd Qu.:89400
Max.   :92700

                        samplingPoint.area     samplingPoint.subArea
SOUTHERN - SOLENT AND SOUTH DOWNS:1968     L&W EAST - SSD :  31
                                           Multiple Values:1937




samplingPoint.samplingPointStatus
CLOSED:   1
OPEN  :1967




                                      samplingPoint.samplingPointType
FRESHWATER - RIVERS                                :968
FRESHWATER - UNSPECIFIED                           :546
GROUNDWATER - BOREHOLE                             :  7
GROUNDWATER - SPRING                               : 17
GROUNDWATER - WELLS & ADITS                        :  1
SALINE WATER - DESIGNATED BATHING BEACHES          :365
SALINE WATER - ESTUARINE SITES - NON BATHING/SHELLFISH: 64
```

```
 phenomenonTime
 Min.   :2020-01-07 10:37:00
 1st Qu.:2022-09-15 04:34:45
 Median :2023-12-13 22:37:30
 Mean   :2023-09-25 03:21:36
 3rd Qu.:2024-12-06 08:53:30
 Max.   :2025-12-19 11:30:00

                                                   samplingPurpose
 ENVIRONMENTAL MONITORING STATUTORY (EU DIRECTIVES)         :689
 MONITORING  (NATIONAL AGENCY POLICY)                       :875
 MONITORING  (UK GOVT POLICY - NOT GQA OR RE)               : 18
 MONITORING (BATHING WATER) SHORT TERM POLLUTION DISREGARDED:  2
 PLANNED INVESTIGATION (LOCAL MONITORING)                   :128
 PLANNED INVESTIGATION (NATIONAL AGENCY POLICY)             :255
 STATUTORY FAILURES (FOLLOW UPS AT NON-DESIGNATED POINTS)   :  1
                    sampleMaterialType    pH Water       Temp Water
 ESTUARINE WATER             :  64     Min.   :6.200   Min.   : 1.90
 GROUNDWATER                 :  25     1st Qu.:7.590   1st Qu.: 9.50
 RIVER / RUNNING SURFACE WATER:1514     Median :7.870   Median :12.70
 SEA WATER                   : 365     Mean   :7.883   Mean   :12.91
                                       3rd Qu.:8.182   3rd Qu.:16.10
                                       Max.   :9.200   Max.   :28.10
                                       NA's   :436     NA's   :25
   Cond @ 25C               location          YEAR          MONTH
 Min.   :   52.8   Isle_of_Wight:1968   Min.   :2020   Min.   : 1.000
 1st Qu.:  480.0                        1st Qu.:2022   1st Qu.: 4.000
 Median :  588.0                        Median :2023   Median : 7.000
 Mean   : 1402.4                        Mean   :2023   Mean   : 6.645
 3rd Qu.:  678.0                        3rd Qu.:2024   3rd Qu.: 9.000
 Max.   :54129.0                        Max.   :2025   Max.   :12.000
 NA's   :447
```

```r
# pH of Water
plot_fun(df, "pH Water")
```

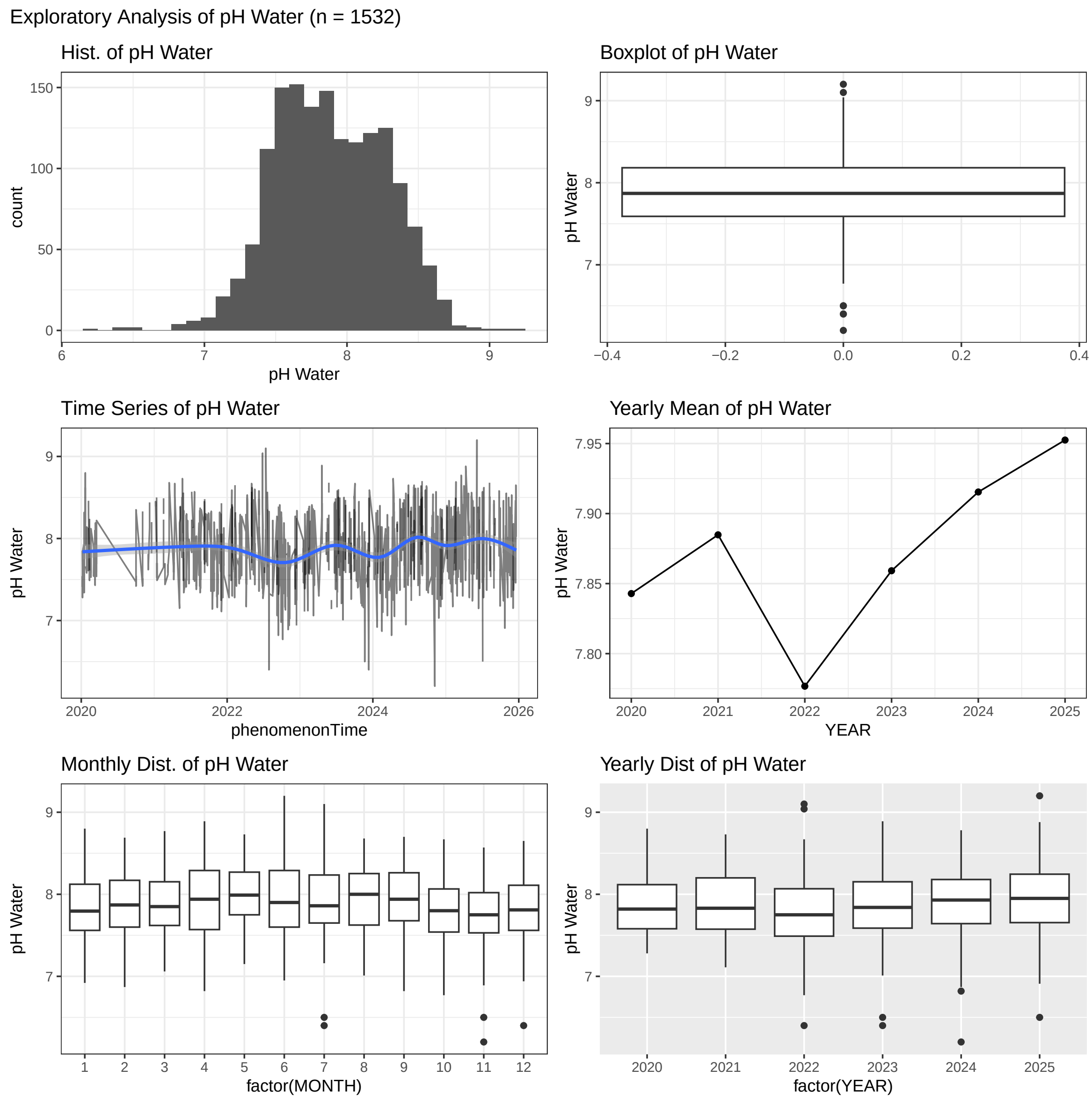

Figure 7: E06000046 Isle of Wight. pH (2020-2025)

```
# Temperature of Water
plot_fun(df, "Temp Water")
```

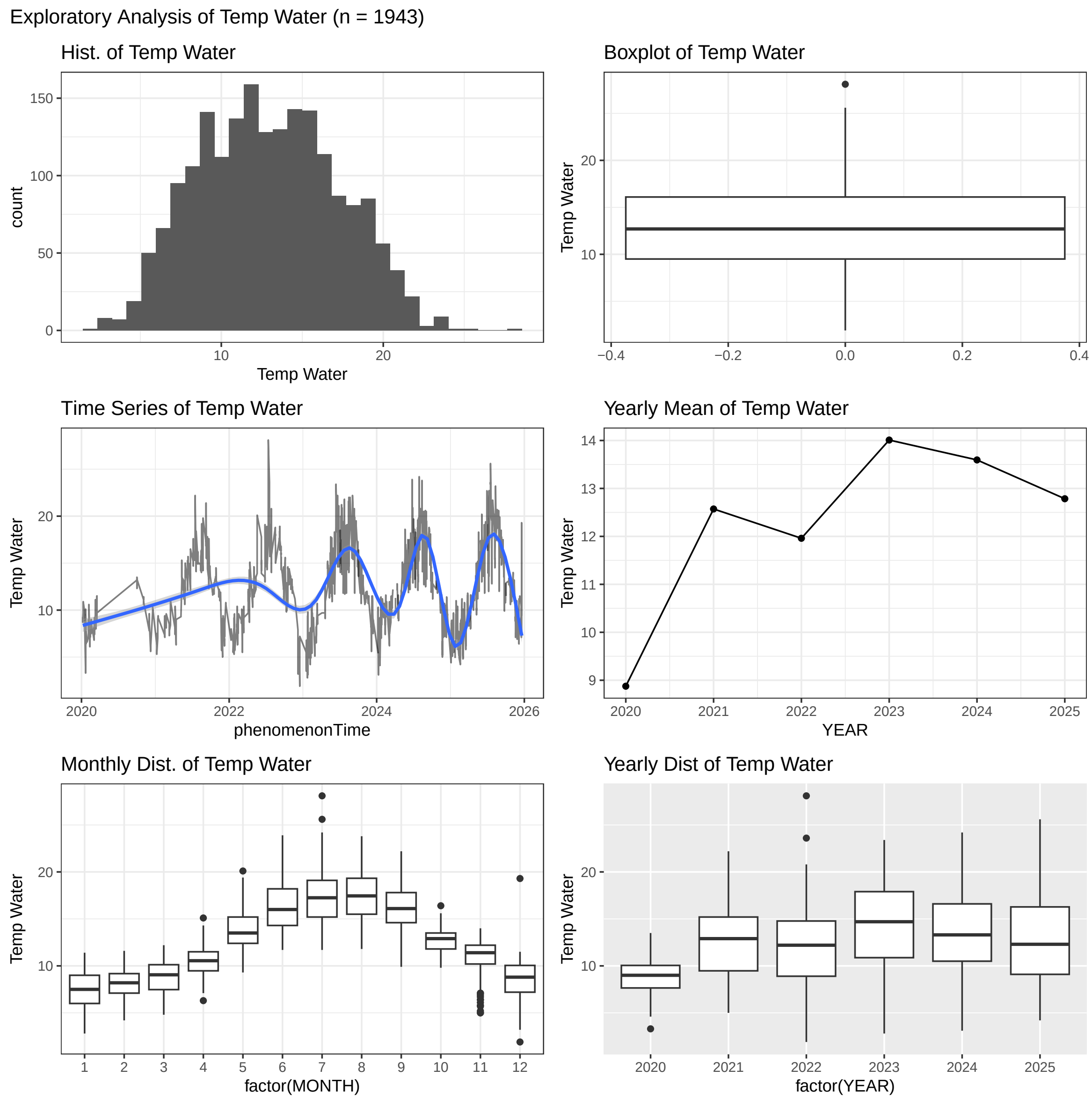

Figure 8: E06000046 Isle of Wight. Temperature (2020-2025)

```
# Conductivity @ 25 degrees
plot_fun(df, "Cond @ 25C")
```

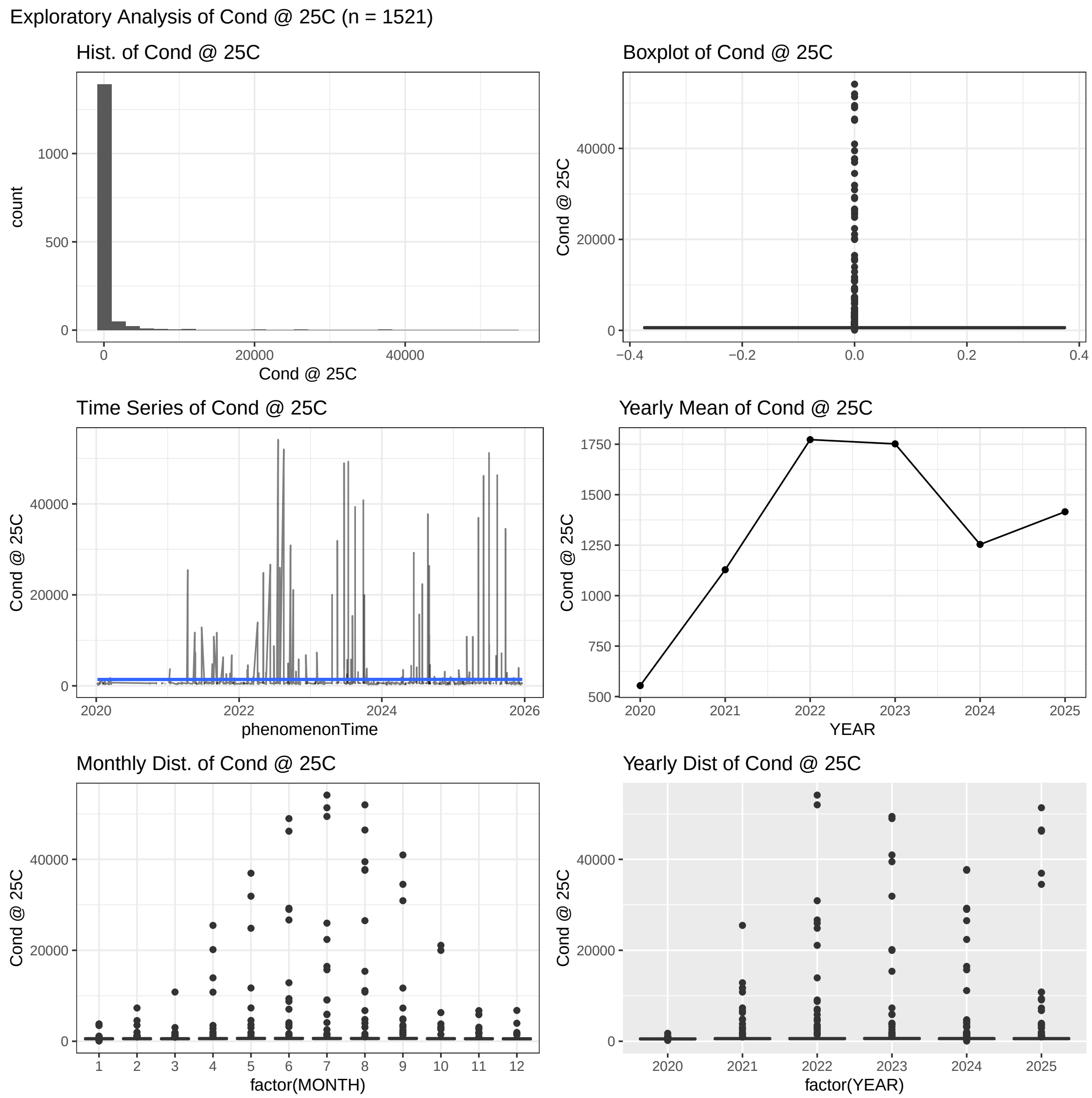

Figure 9: E06000046 Isle of Wight. Conductivity @ 25C (2020-2025)

E07000091_New_Forest

```
# ------ 1. Data cleaning and quality check ------

# Specify the dataset
```

```r
df <- nfo # <- only change this

# categorical nominal variables as factors
df$location <- as.factor(df$location)
df$samplingPoint.notation <- as.factor(df$samplingPoint.notation)
df$samplingPoint.prefLabel <- as.factor(df$samplingPoint.prefLabel)
df$samplingPoint.region <- as.factor(df$samplingPoint.region)
df$samplingPoint.area <- as.factor(df$samplingPoint.area)
df$samplingPoint.subArea <- as.factor(df$samplingPoint.subArea)
df$samplingPoint.samplingPointStatus <- as.factor(df$samplingPoint.samplingPointStatus)
df$samplingPoint.samplingPointType <- as.factor(df$samplingPoint.samplingPointType)
df$samplingPurpose <- as.factor(df$samplingPurpose)
df$sampleMaterialType <- as.factor(df$sampleMaterialType)

# create month end year
df <- df %>%
  mutate(
    YEAR = year(phenomenonTime),
    MONTH = month(phenomenonTime)
  )

# check for missing data
colSums(is.na(df))
```

```
                             Date            samplingPoint.notation
                                0                                 0
          samplingPoint.prefLabel             samplingPoint.easting
                                0                                 0
           samplingPoint.northing              samplingPoint.region
                                0                                 0
               samplingPoint.area             samplingPoint.subArea
                                0                                 0
samplingPoint.samplingPointStatus   samplingPoint.samplingPointType
                                0                                 0
                   phenomenonTime                   samplingPurpose
                                0                                 0
               sampleMaterialType                          pH Water
                                0                               384
                       Temp Water                       Cond @ 25C
                              583                               915
                         location                              YEAR
                                0                                 0
```

```
                                  MONTH
                                      0
```

```r
# remove duplicates, if any
df <- df %>%
  distinct()

# check structure
str(df)
```

```
'data.frame':   3462 obs. of  19 variables:
 $ Date                            : IDate, format: "2020-02-19" "2021-04-
29" ...
 $ samplingPoint.notation          : Factor w/ 120 levels "SO-5GWQ0378",..: 93 93 93 93 93 9
 $ samplingPoint.prefLabel         : Factor w/ 120 levels "20M U/S SHEPTON BRIDGE, SHEPTON W
 $ samplingPoint.easting           : int  413550 413550 413550 413550 413550 413550 413550 4
 $ samplingPoint.northing          : int  114160 114160 114160 114160 114160 114160 114160 1
 $ samplingPoint.region            : Factor w/ 2 levels "Southern","SouthWest": 2 2 2 2 2 2
 $ samplingPoint.area              : Factor w/ 2 levels "SOUTH WEST - WESSEX",..: 1 1 1 1 1
 $ samplingPoint.subArea           : Factor w/ 3 levels "EAST WESSEX",..: 1 1 1 1 1 1 1 1 1
 $ samplingPoint.samplingPointStatus: Factor w/ 1 level "OPEN": 1 1 1 1 1 1 1 1 1 1 ...
 $ samplingPoint.samplingPointType  : Factor w/ 17 levels "AGRICULTURE - FISH FARMING - NOT W
 $ phenomenonTime                  : POSIXct, format: "2020-02-19 09:43:00" "2021-
04-29 11:16:00" ...
 $ samplingPurpose                 : Factor w/ 9 levels "COMPLIANCE AUDIT (PERMIT)",..: 1 1
 $ sampleMaterialType              : Factor w/ 9 levels "ANY TRADE EFFLUENT",..: 9 9 9 9 9 9
 $ pH Water                        : num  7.94 8.17 8.04 7.91 7.75 7.91 7.96 7.89 8 8.01 ..
 $ Temp Water                      : num  NA NA NA NA NA NA NA NA NA NA ...
 $ Cond @ 25C                      : num  NA NA NA NA NA NA NA NA NA NA ...
 $ location                        : Factor w/ 1 level "New_Forest": 1 1 1 1 1 1 1 1 1 1 ..
 $ YEAR                            : num  2020 2021 2021 2021 2021 ...
 $ MONTH                           : num  2 4 5 6 7 8 9 10 11 12 ...
```

```r
# check ranges
summary(df)
```

```
      Date              samplingPoint.notation
 Min.   :2020-01-06   SO-G0004128: 243
 1st Qu.:2022-04-05   SW-50280585: 237
 Median :2023-10-24   SW-50281314: 158
 Mean   :2023-07-28   SW-50281306:  83
```

```
 3rd Qu.:2024-12-11   SW-50280458:  72  
 Max.   :2025-12-23   SO-G0004187:  71  
                      (Other)    :2598  
                             samplingPoint.prefLabel samplingPoint.easting
 DARK WATER AT GATEWOOD BRIDGE           : 243       Min.   :408700       
 HAMPSHIRE AVON AT FORDINGBRIDGE         : 237       1st Qu.:415748       
 DOCKENS WATER AT A338                   : 158       Median :430225       
 DOCKENS  WATER AT MOYLES COURT FOOTBRIDGE:  83       Mean   :427556       
 RINGWOOD STW FINAL EFFLUENT             :  72       3rd Qu.:435603       
 BROCKENHURST SEWAGE TREATMENT WORKS     :  71       Max.   :448138       
 (Other)                                 :2598                            
 samplingPoint.northing samplingPoint.region
 Min.   : 91803         Southern :2282      
 1st Qu.:101417         SouthWest:1180      
 Median :107092                             
 Mean   :106442                             
 3rd Qu.:113150                             
 Max.   :118649                             

                        samplingPoint.area     samplingPoint.subArea
 SOUTH WEST - WESSEX             :1180     EAST WESSEX    :1180     
 SOUTHERN - SOLENT AND SOUTH DOWNS:2282     L&W EAST - SSD :  16     
                                           Multiple Values:2266     




 samplingPoint.samplingPointStatus
 OPEN:3462                        






                                         samplingPoint.samplingPointType
 FRESHWATER - RIVERS                                    :2457            
 SEWAGE DISCHARGES - FINAL/TREATED EFFLUENT - WATER COMPANY: 214            
 FRESHWATER - UNSPECIFIED                               : 201            
 AGRICULTURE - FISH FARMING - NOT WATER COMPANY         : 163            
 FRESHWATER - COMPARATIVE INLET POINTS                  :  98            
 SALINE WATER - DESIGNATED BATHING BEACHES              :  60            
 (Other)                                                : 269            
```

```
phenomenonTime
Min.   :2020-01-06 11:59:00
1st Qu.:2022-04-05 11:41:15
Median :2023-10-24 22:35:30
Mean   :2023-07-29 03:47:45
3rd Qu.:2024-12-11 12:09:00
Max.   :2025-12-23 13:23:00

                                                      samplingPurpose
ENVIRONMENTAL MONITORING STATUTORY (EU DIRECTIVES)    :1024
MONITORING  (NATIONAL AGENCY POLICY)                  : 985
MONITORING  (UK GOVT POLICY - NOT GQA OR RE)          : 444
COMPLIANCE AUDIT (PERMIT)                             : 335
PLANNED INVESTIGATION (LOCAL MONITORING)              : 325
WATER QUALITY OPERATOR SELF MONITORING COMPLIANCE DATA: 257
(Other)                                               :  92
                                      sampleMaterialType    pH Water
RIVER / RUNNING SURFACE WATER                  :2785     Min.   : 5.540
FINAL SEWAGE EFFLUENT                          : 318     1st Qu.: 7.060
TRADE EFFLUENT - FRESHWATER RETURNED ABSTRACTED: 163     Median : 7.410
SEA WATER                                      :  60     Mean   : 7.417
POND / LAKE / RESERVOIR WATER                  :  49     3rd Qu.: 7.800
GROUNDWATER - PURGED/PUMPED/REFILLED           :  27     Max.   :10.100
(Other)                                        :  60     NA's   :384
  Temp Water      Cond @ 25C           location         YEAR
Min.   : 0.30   Min.   :   64.0   New_Forest:3462   Min.   :2020
1st Qu.: 8.20   1st Qu.:  152.0                     1st Qu.:2022
Median :11.90   Median :  236.0                     Median :2023
Mean   :11.72   Mean   :  352.3                     Mean   :2023
3rd Qu.:15.30   3rd Qu.:  422.0                     3rd Qu.:2024
Max.   :27.90   Max.   :44310.0                     Max.   :2025
NA's   :583     NA's   :915
    MONTH
Min.   : 1.000
1st Qu.: 4.000
Median : 7.000
Mean   : 6.623
3rd Qu.:10.000
Max.   :12.000
```

```r
# pH of Water
plot_fun(df, "pH Water")
```

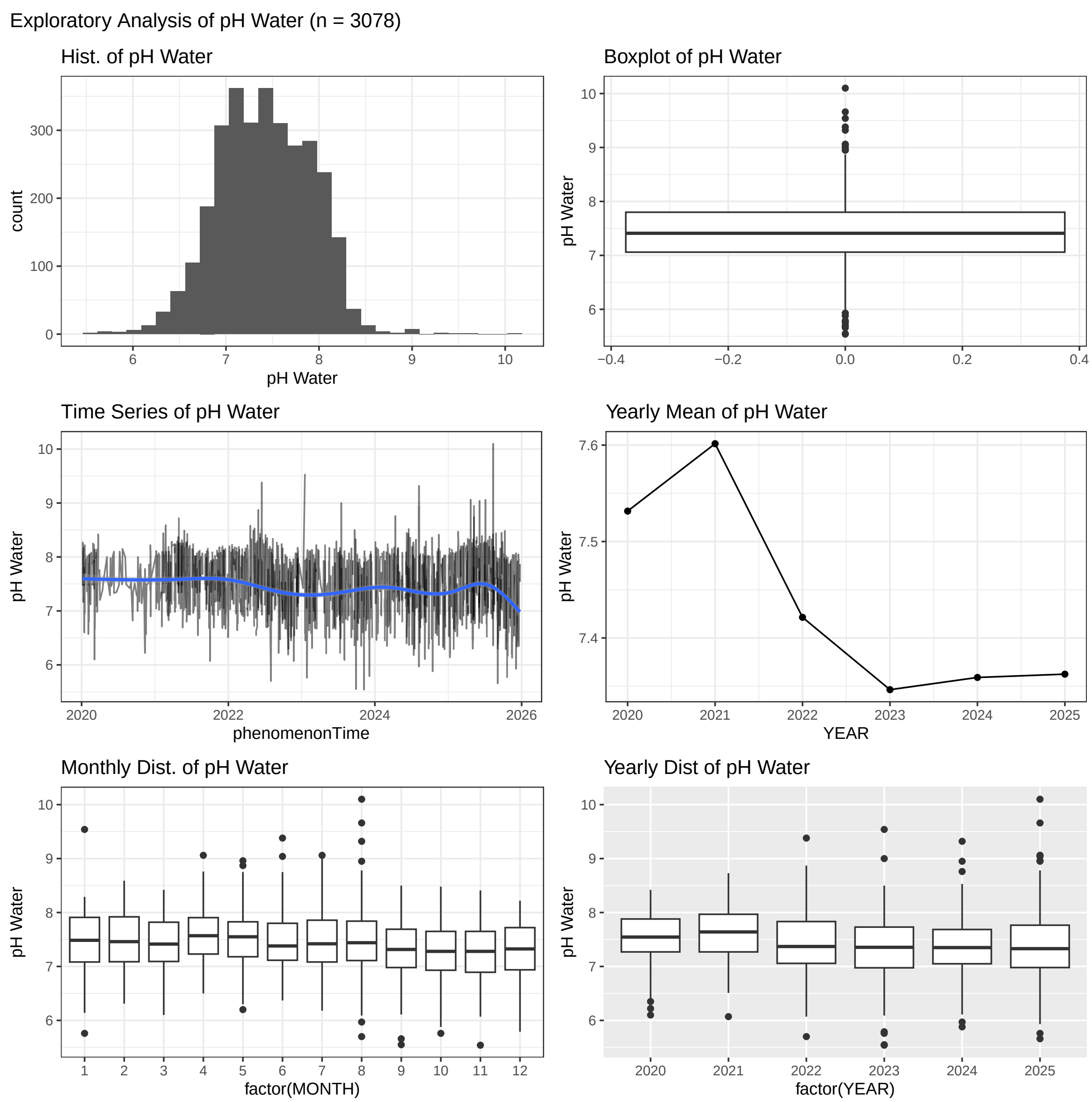

Figure 10: E07000091 New Forest. pH (2020-2025)

```r
# Temperature of Water
plot_fun(df, "Temp Water")
```

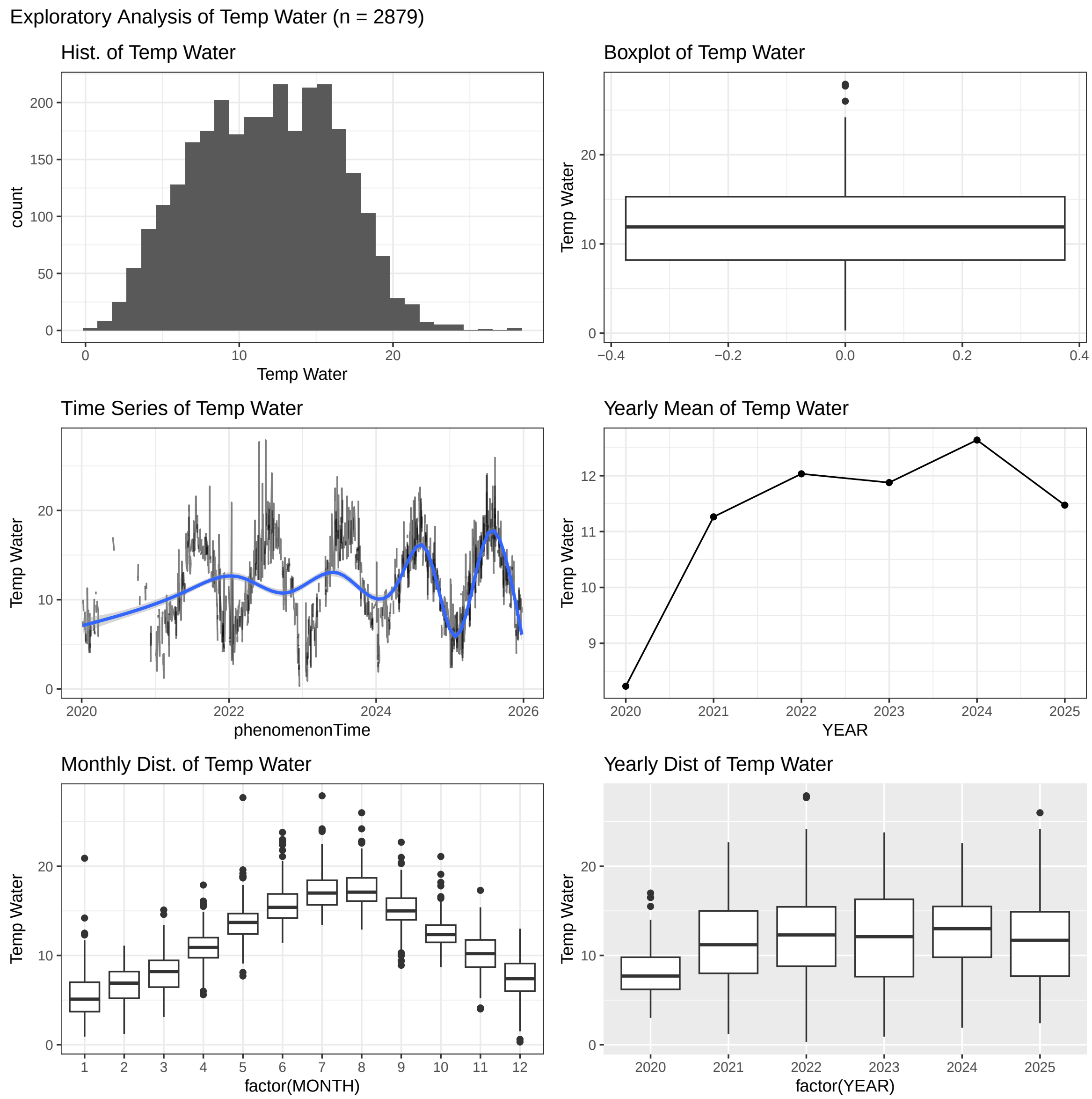

Figure 11: E07000091 New Forest. Temperature (2020-2025)

```r
# Conductivity @ 25 degrees
plot_fun(df, "Cond @ 25C")
```

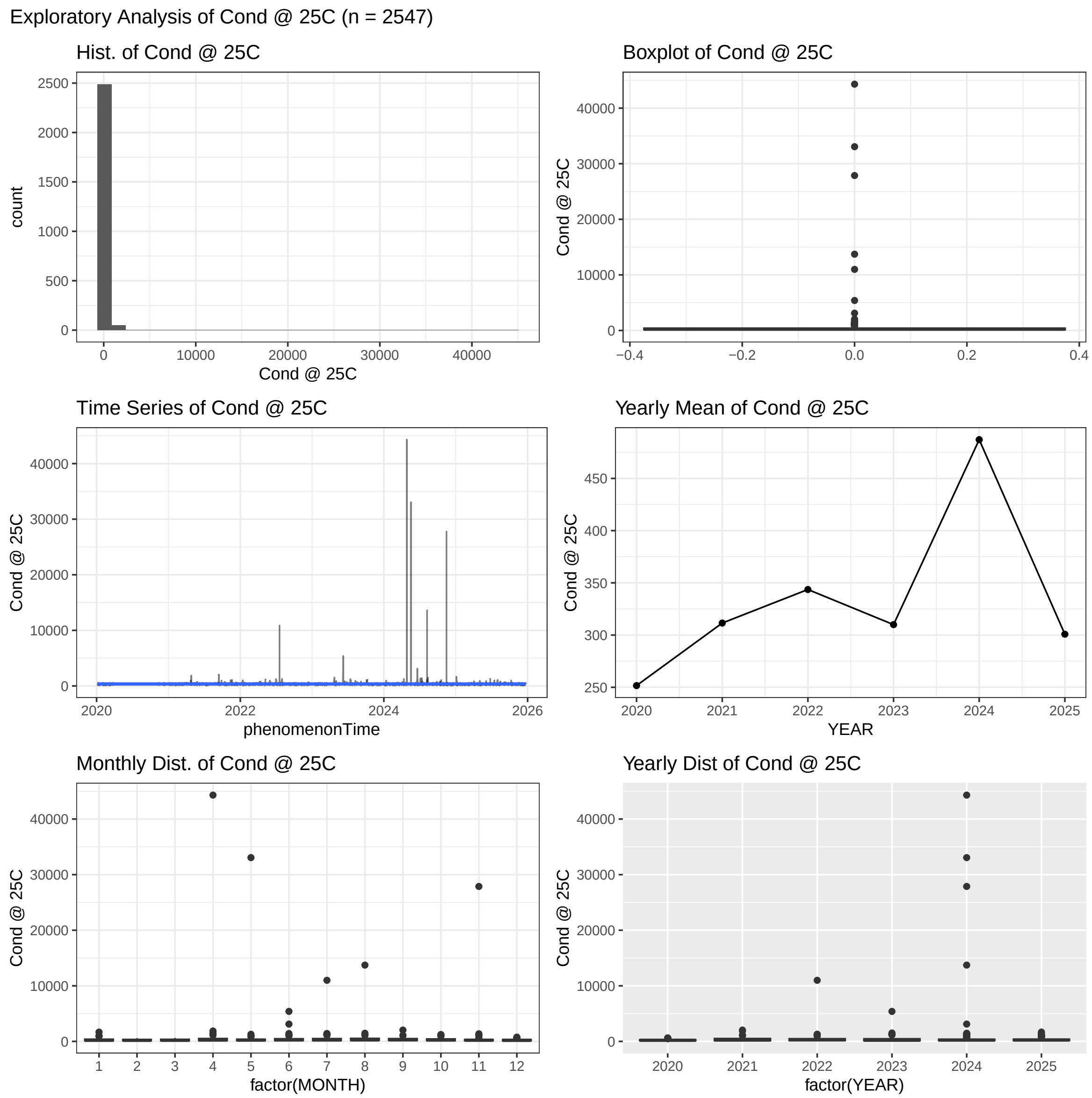

Figure 12: E07000091 New Forest. Conductivity @ 25C (2020-2025)

Comperative analysis

```
# combine four datasets
df <- bind_rows(med, rot, iow, nfo)
```

```r
# categorical nominal variables as factors
df$location <- as.factor(df$location)
df$samplingPoint.notation <- as.factor(df$samplingPoint.notation)
df$samplingPoint.prefLabel <- as.factor(df$samplingPoint.prefLabel)
df$samplingPoint.region <- as.factor(df$samplingPoint.region)
df$samplingPoint.area <- as.factor(df$samplingPoint.area)
df$samplingPoint.subArea <- as.factor(df$samplingPoint.subArea)
df$samplingPoint.samplingPointStatus <- as.factor(df$samplingPoint.samplingPointStatus)
df$samplingPoint.samplingPointType <- as.factor(df$samplingPoint.samplingPointType)
df$samplingPurpose <- as.factor(df$samplingPurpose)
df$sampleMaterialType <- as.factor(df$sampleMaterialType)

# create month end year
df <- df %>%
  mutate(
    YEAR = year(phenomenonTime),
    MONTH = month(phenomenonTime)
  )

# view sampleMaterialType per location
table(df$sampleMaterialType, df$location)
```

```
                                               Isle_of_Wight Medway
  ANY TRADE EFFLUENT                                       0      0
  ESTUARINE WATER                                         64    137
  FINAL SEWAGE EFFLUENT                                    0      0
  GROUNDWATER                                             25      5
  GROUNDWATER - PURGED/PUMPED/REFILLED                     0      0
  POND / LAKE / RESERVOIR WATER                            0     57
  RIVER / RUNNING SURFACE WATER                         1514     16
  SEA WATER                                              365      0
  TRADE EFFLUENT - FRESHWATER RETURNED ABSTRACTED          0      0

                                               New_Forest Rother
  ANY TRADE EFFLUENT                                    22     50
  ESTUARINE WATER                                       14      0
  FINAL SEWAGE EFFLUENT                                318    383
  GROUNDWATER                                           24     20
  GROUNDWATER - PURGED/PUMPED/REFILLED                  27      0
  POND / LAKE / RESERVOIR WATER                         49     64
```

```
  RIVER / RUNNING SURFACE WATER                         2785   1352
  SEA WATER                                               60     65
  TRADE EFFLUENT - FRESHWATER RETURNED ABSTRACTED        163      0
```

```
# view variables and observations
str(df)
```

```
'data.frame':   7579 obs. of  19 variables:
 $ Date                             : IDate, format: "2020-01-10" "2020-02-
25" ...
 $ samplingPoint.notation           : Factor w/ 274 levels "SO-5GWQ0378",..: 72 72 72 72 72 
 $ samplingPoint.prefLabel          : Factor w/ 274 levels "20M U/S SHEPTON BRIDGE, SHEPTON 
 $ samplingPoint.easting            : int  572949 572949 572949 572949 572949 572949 572949 
 $ samplingPoint.northing           : int  176722 176722 176722 176722 176722 176722 176722 
 $ samplingPoint.region             : Factor w/ 3 levels "Southern","SouthWest",..: 1 1 1 1 
 $ samplingPoint.area               : Factor w/ 4 levels "SOUTH WEST - WESSEX",..: 2 2 2 2 2
 $ samplingPoint.subArea            : Factor w/ 6 levels "EAST WESSEX",..: 3 3 3 3 3 3 3 3 3
 $ samplingPoint.samplingPointStatus: Factor w/ 2 levels "CLOSED","OPEN": 2 2 2 2 2 2 2 2 2 
 $ samplingPoint.samplingPointType  : Factor w/ 21 levels "AGRICULTURE - FISH FARMING - NOT 
 $ phenomenonTime                   : POSIXct, format: "2020-01-10 09:38:00" "2020-
02-25 12:01:00" ...
 $ samplingPurpose                  : Factor w/ 13 levels "COMPLIANCE AUDIT (PERMIT)",..: 1 
 $ sampleMaterialType               : Factor w/ 9 levels "ANY TRADE EFFLUENT",..: 2 2 2 2 2 
 $ pH Water                         : num  8.45 8.6 8.63 8.43 8.74 8.67 8.61 8.69 8.83 8.66 
 $ Temp Water                       : num  NA NA NA NA NA NA NA NA NA NA ...
 $ Cond @ 25C                       : num  44870 40656 38052 41183 40625 ...
 $ location                         : Factor w/ 4 levels "Isle_of_Wight",..: 2 2 2 2 2 2 2 2
 $ YEAR                             : num  2020 2020 2020 2021 2021 ...
 $ MONTH                            : num  1 2 3 10 11 12 1 2 3 5 ...
```

```
# Create broader categories for samplingPointType
df <- df %>%
  mutate(
    sampleMaterialGroup = fct_collapse(
      sampleMaterialType,
      Freshwater = c("RIVER / RUNNING SURFACE WATER", "POND / LAKE / RESERVOIR WATER", "ESTU
      Groundwater = c("GROUNDWATER - PURGED/PUMPED/REFILLED", "GROUNDWATER"),
      Sewage = c("FINAL SEWAGE EFFLUENT"),
      Seawater = c("SEA WATER")
    )
  )
```

```r
# Means on non-missing values (pH and temp only)
df <- df %>%
  group_by(location,sampleMaterialGroup) %>%
  summarise(
    mean_temp = mean(`Temp Water`, na.rm = TRUE),
    mean_ph = mean(`pH Water`, na.rm = TRUE),
    mean_cond = mean(`Cond @ 25C`, na.rm = TRUE),
    .groups = "drop"
  )

# quick view variables and observations
str(df)
```

```
tibble [13 x 5] (S3: tbl_df/tbl/data.frame)
 $ location           : Factor w/ 4 levels "Isle_of_Wight",..: 1 1 1 2 2 3 3 3 3 4 ...
 $ sampleMaterialGroup: Factor w/ 4 levels "Freshwater","Sewage",..: 1 3 4 1 3 1 2 3 4 1 ...
 $ mean_temp          : num [1:13] 11.6 12.3 18.4 12.1 12.8 ...
 $ mean_ph            : num [1:13] 7.89 7.18 NaN 8.43 6.96 ...
 $ mean_cond          : num [1:13] 1414 623 NaN 28898 1213 ...
```

Sample Material Type and Location

pH

```r
# compare mean pH by location
ggplot(df,
       aes(x = location, y = mean_ph, fill = location)) +
  geom_col() +
  facet_wrap(~ sampleMaterialGroup) +
  labs(
    x = "Local authority area",
    y = "Mean water pH",
    fill = "Local authority area",
    title = "Mean pH by Site Type and Location") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "right")
```

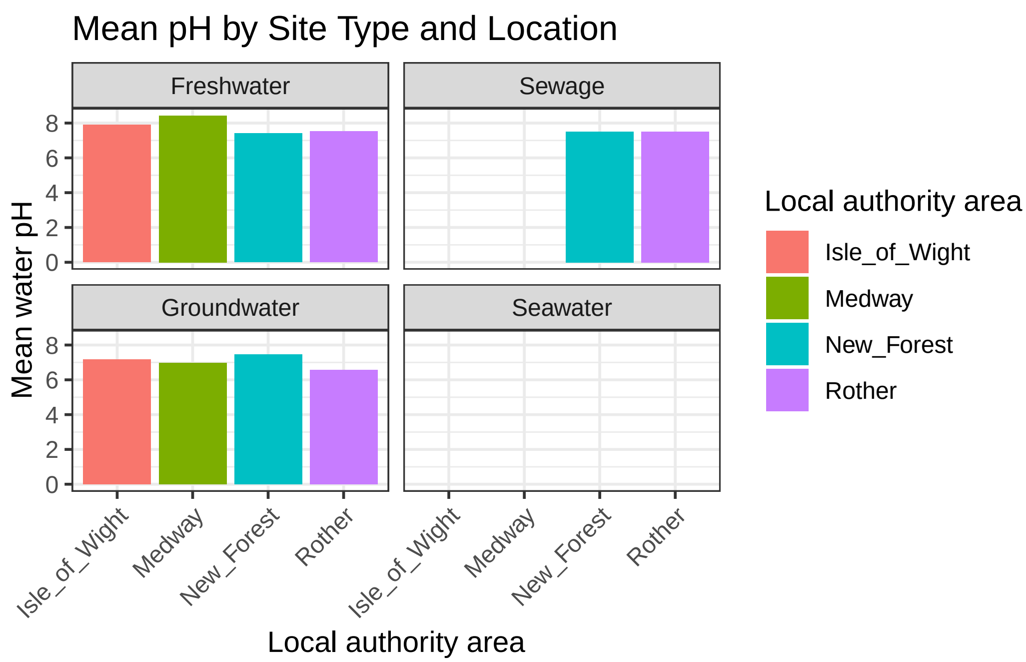

Figure 13: Mean pH by Sample Material Type and Location (2020-2025)

Temperature

```
# compare mean temperature by location
ggplot(df,
       aes(x = location, y = mean_temp, fill = location)) +
  geom_col() +
  facet_wrap(~ sampleMaterialGroup) +
  labs(
    x = "Local authority area",
    y = "Mean water temepráture",
    fill = "Local authority area",
    title = "Mean Temerpature by Site Type and Location") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "right")
```

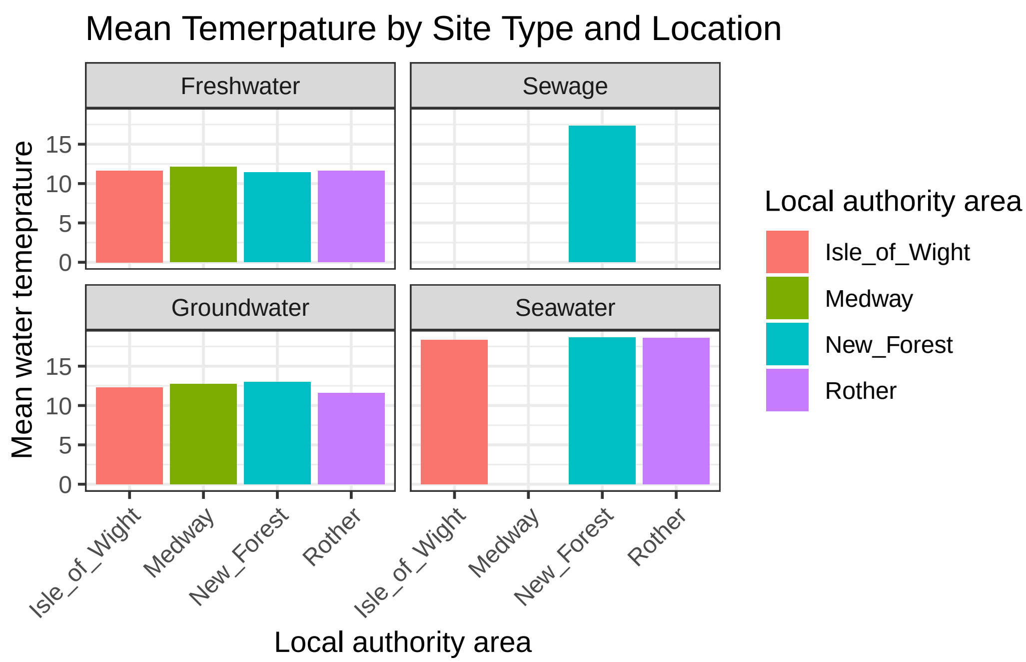

Figure 14: Mean Temerpature by Sample Material Type and Location (2020-2025)

Conductivity @ 25C

```
ggplot(df,
       aes(x = location, y = mean_cond, fill = location)) +
  geom_col() +
  facet_wrap(~ sampleMaterialGroup) +
  labs(
    x = "Local authority area",
    y = "Mean water conductivity @ 25C",
    fill = "Local authority area",
    title = "Mean Conductivity by Site Type and Location") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "right")
```

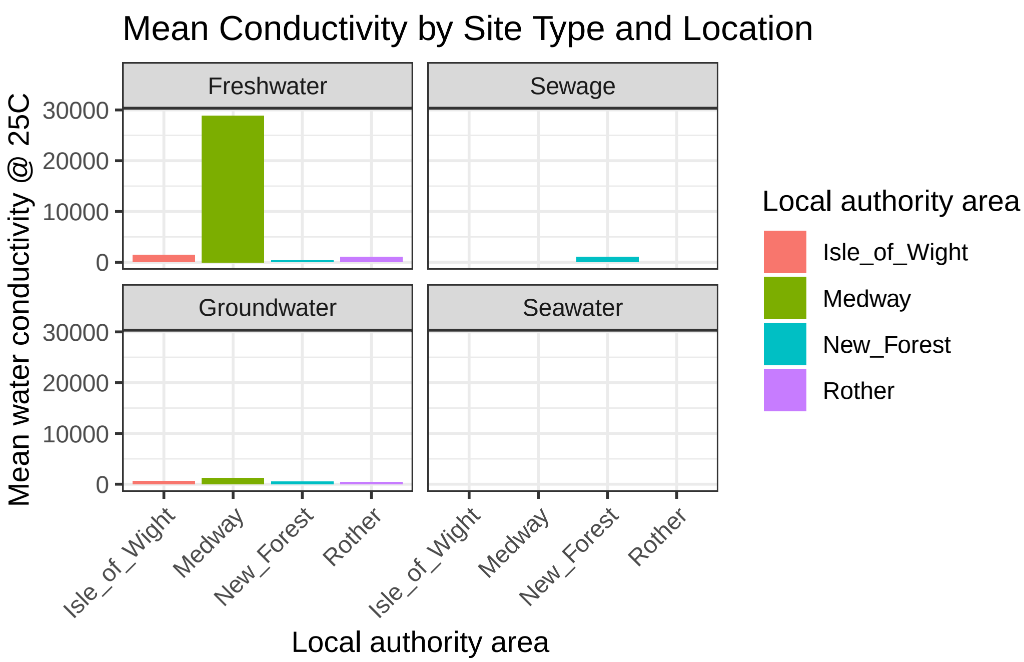

Figure 15: Mean Conductivity by Sample Material Type and Location (2020-2025)